



Friction Management

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B O S T O N

Results of Uncontrolled Friction: In the Public



While the public may not specifically think about friction... its effects are present!

Train Noise



Train Noise in the Media



Ask ChatGPT for transit noise news articles... and there is no shortage of news!

1. "Which Tube line is loudest? We rank the noisiest Underground sections" — *The Times*, late July 2026
Tests found noise exceeding 100 dB on several London Underground segments. The article examines rail corrugation, passenger and driver exposure, complaints, and Transport for London's maintenance response. The Times
2. "High-speed trains would get new noise regulations in FRA proposal" — *Smart Cities Dive*, August 3, 2026
Covers a proposed U.S. regulation creating alternative noise standards for trains operating above 160 mph, intended to facilitate high-speed rail while retaining public-health protections. Smart Cities Dive +1
3. "FRA Proposes Updated Noise Standards to Support US High-Speed Rail" — *Railway-News*, August 3, 2026
An industry-focused account of the same Federal Railroad Administration proposal, including its implications for future high-speed passenger-rail projects and technology approvals. Railway-News
4. "West-end Toronto residents frustrated with Metrolinx over noise protections" — *CityNews Toronto*, July 14, 2026
Residents near the planned Bloor–Lansdowne GO station are calling for noise walls and other protections as service and infrastructure expand along the Barrie rail corridor. CityNews Toronto
5. "Bellevue residents say new cross-lake rail line violates noise limits" — *KOMO News*, June 2026
Homeowners near Sound Transit's 2 Line allege that a low-frequency rumble from trains crossing the East Channel Bridge disrupts sleep and may exceed Bellevue's noise limits. KOMO
6. "Enatai Residents Threaten Legal Action Against Sound Transit Over 2 Line Noise" — *Downtown Bellevue Network*, June 15, 2026
Reports residents' threatened lawsuit, overnight train observations and concerns about noise generated when trains pass over a bridge expansion joint. Downtown Belle...

7. "East Toronto residents near Lakeshore East train line call on Metrolinx to implement noise and vibration mitigation measures" — *Beach Metro Community News*, May 28, 2026
Discusses neighborhood concerns that an additional GO Transit track and increased service will produce more noise and vibration without sufficient mitigation. Beach Metro Co...
8. "Transportation Noise: An Overlooked Risk to Heart Health" — American College of Cardiology, March 23, 2026
Summarizes research connecting transportation noise with cardiovascular risk; the reported analysis associated loud rail noise with a higher risk of adverse cardiovascular events. American Colleg...
9. "Suburban Rail Loop construction impacts health of Heatherton residents" — *Herald Sun*, April 2026
Focuses on continuous construction noise and vibration around Melbourne's Suburban Rail Loop stabling-yard project, including complaints about sleep disruption and mitigation measures. Herald Sun
10. "Ad-weary New Yorkers school MTA on 'annoying' 75-decibel subway commercials" — *New York Post*, February 26, 2026
Covers rider objections to audio advertising in New York subway and commuter-rail stations, framing the advertisements as an additional source of transit-system noise pollution. New York Post
11. "West-end quiet zone construction delayed in Beaumont" — *Beaumont Enterprise*, February 2026
Reports on work to restore a railroad quiet zone by upgrading crossings so routine horn sounding can be reduced—a community-noise issue involving passenger and freight rail operations. Beaumont Enter...
12. "Managing Rail Noise and Vibration on Both Sides of the Pond" — *RWDI*, July 2026
A professional-industry recap of an APTA Rail 2026 discussion comparing U.S. and U.K. approaches to measuring, regulating and mitigating transit-rail noise and vibration. rwdi.com

Results of Uncontrolled Friction: In the Public



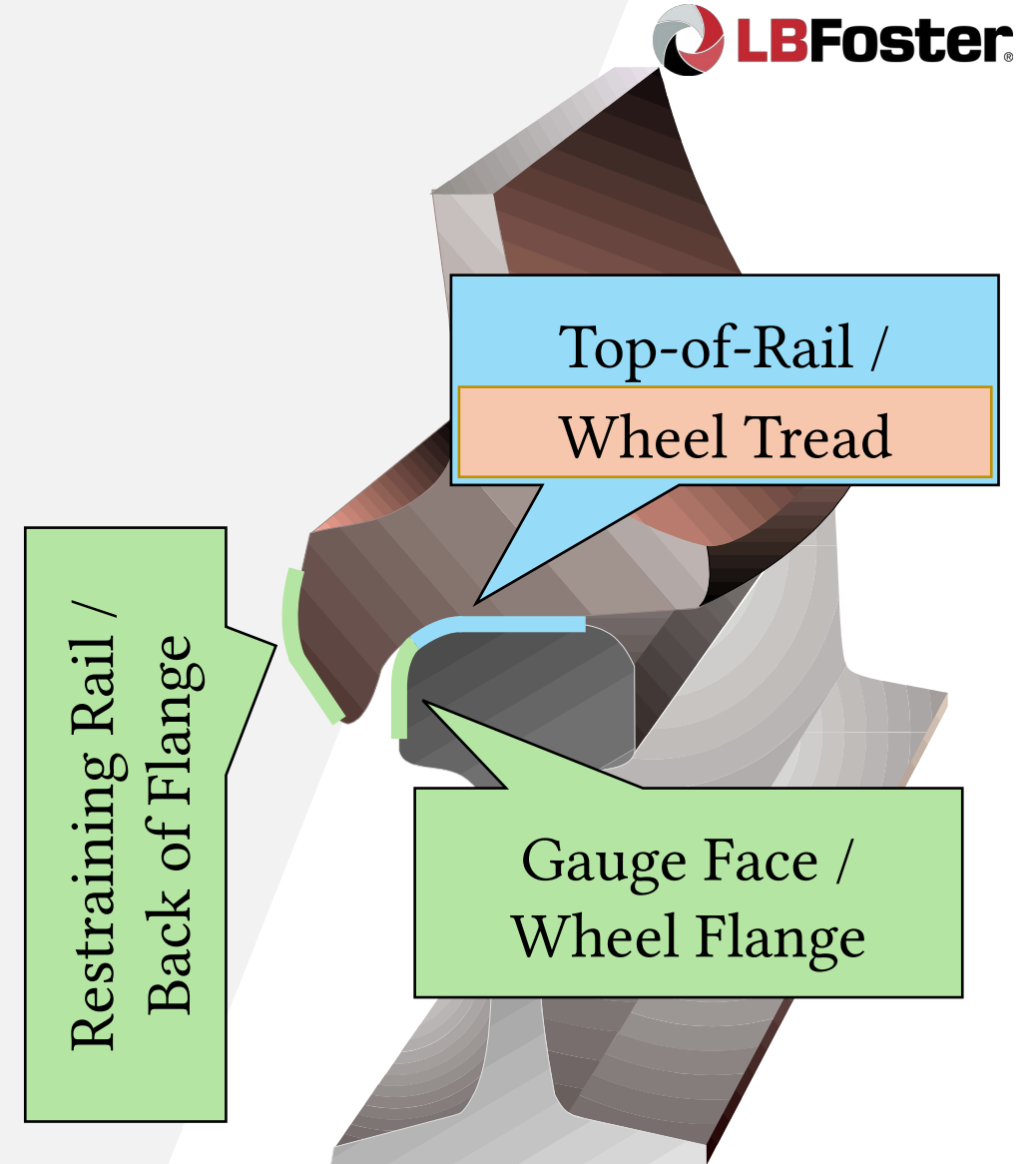
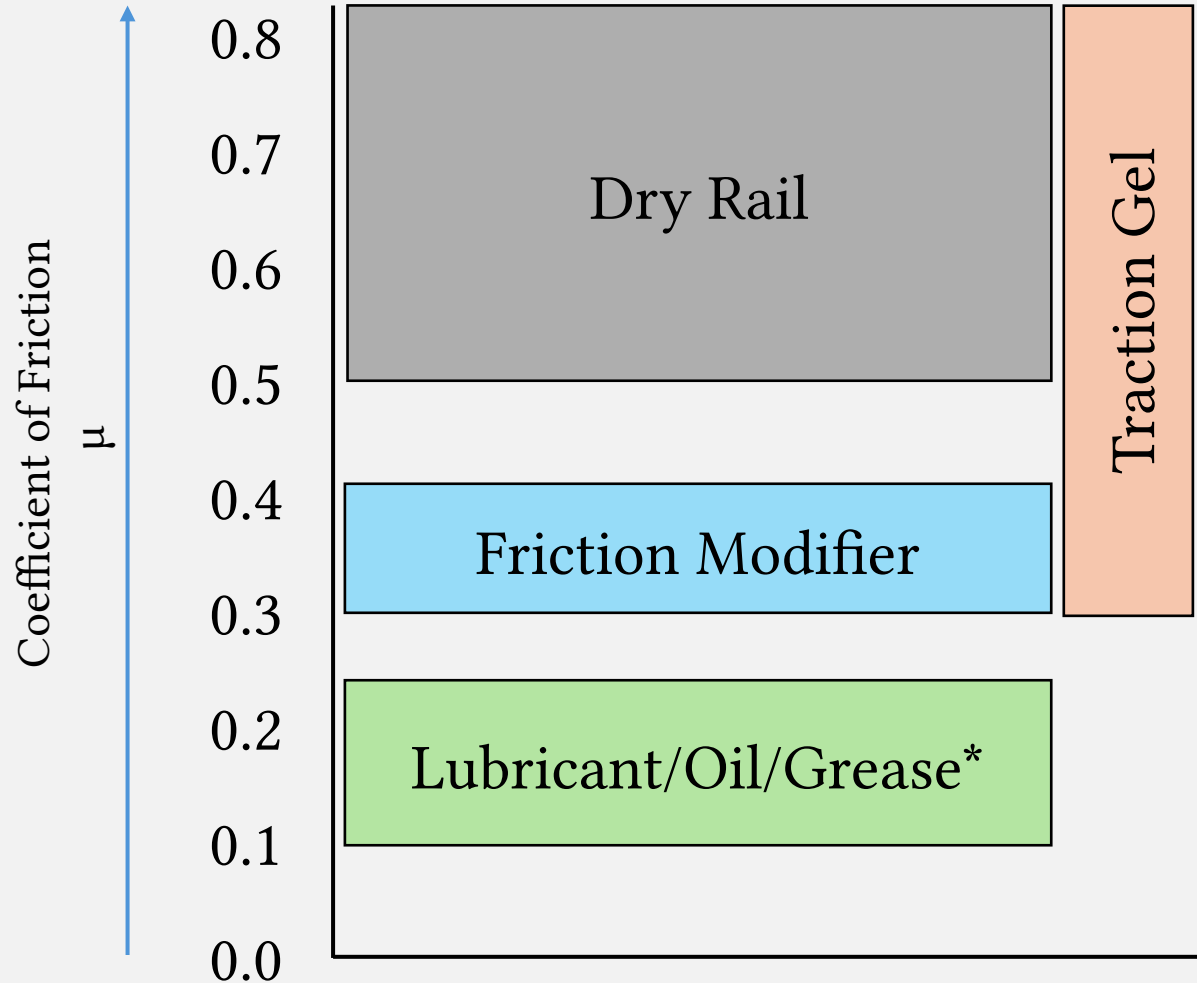
- Service Disruptions and Interruptions
- Delayed Opening of New Transit Lines

Subway train derails in Massachusetts and injures some riders



Noise and many aspects of service disruptions can often be related to the friction level!

What is Controlled Friction?



Outline

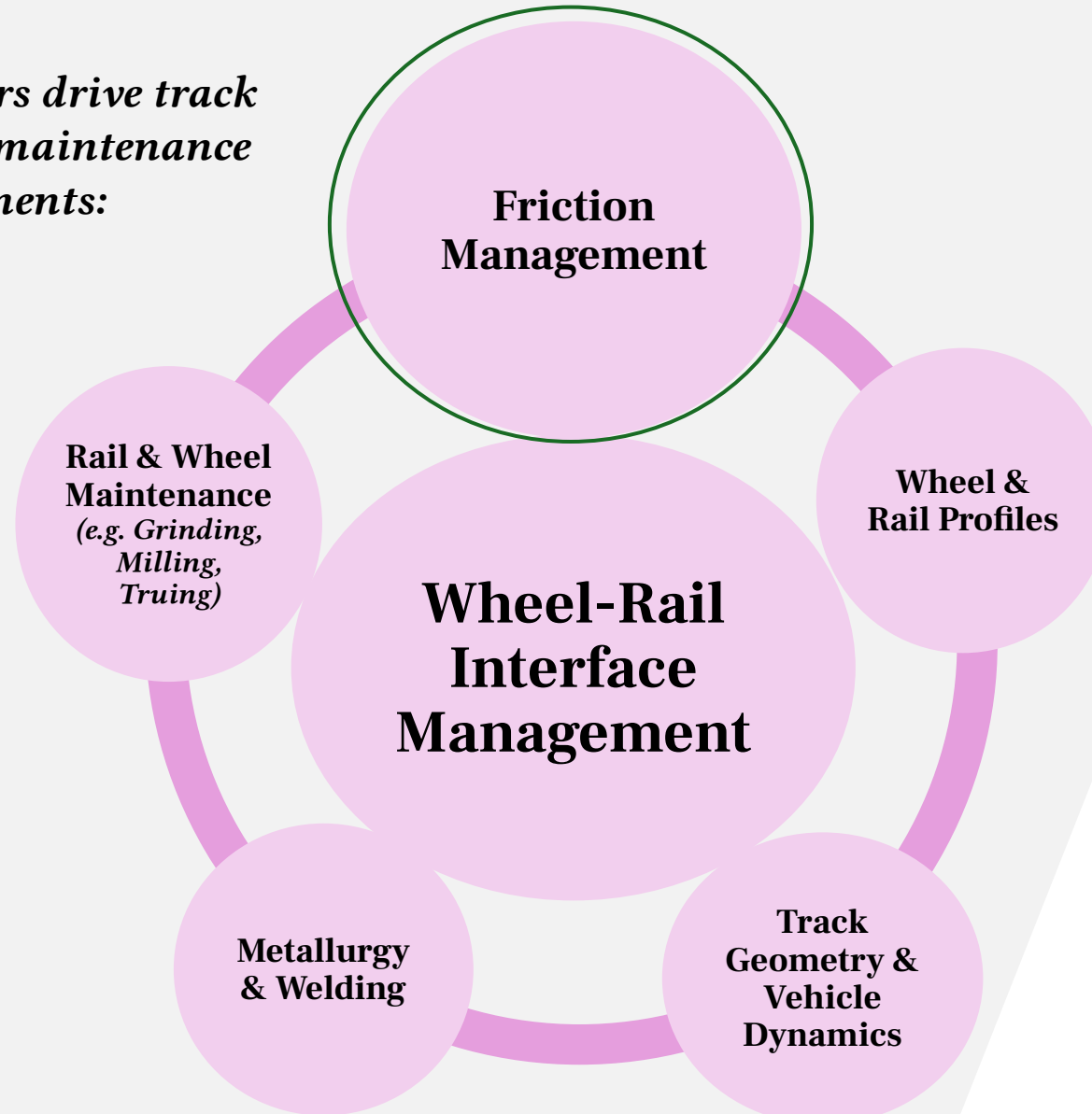


- **Friction Management Introduction**
- **Top-of-Rail / Wheel Tread Application**
- **Gauge Face / Restraining Rail / Wheel Flange Application**
- **Validation & Measurement**
- **Maintenance**

Friction Management



Five main factors drive track reliability and maintenance requirements:



Trains will always face frictional challenges, especially in curves! Transits also have lots of unique track designs, vehicle types, etc. that impose additional challenges.

Friction Management directly improves track reliability, reduces maintenance interventions, and starts protecting your infrastructure immediately, regardless of age and condition.

Tangible Benefits of Friction Management



- Long history of well-documented publicly available results from many conferences: WRI, ICRI, APTA, CM, AREMA, etc.

Immediate Benefits	Longer Term Benefits
Reduce Wear (~25-60%)	Reduce Grinding Intervals (~50%)
Reduce Noise (~5-10dBA)	Reduce Defects (~50%)
Enhance Safety (Nadal/Weinstock Limit)	Reduce Fastener Damage (~20-80%)
Improve Energy Consumption (~3-5%)	Improve Ride Quality (Vehicle Steering, Reduce Hunting)
Payback / Return on Investment (< 1 year)	

Broadly speaking, savings will be different for every system!

Choosing a Friction Management Solution



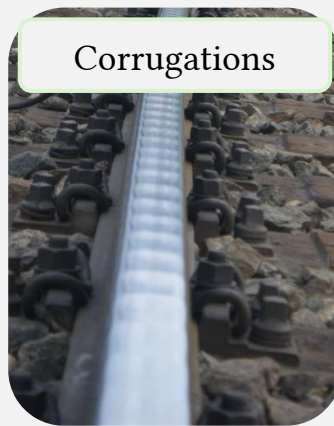
What is the most pertinent issue(s)?

There could be several things!

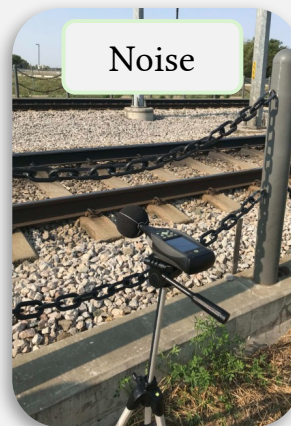
Rolling Contact Fatigue



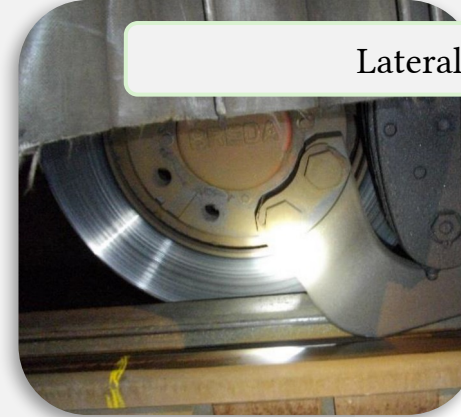
Corrugations



Noise



Lateral Forces



Running Rail Wear



Restraining Rail Wear



Wheel Wear



Traction



- Must be cognizant of budget constraints for most transit agencies
- Speak with a supplier or consultant who specializes in Friction Management

End Goals of FM Implementation



- Correct equipment and consumable solution implemented for the targeted issues
- Choosing products that are backed by proven data and results
 - Be careful of marketing information, especially in the modern world of AI!
 - When applied correctly, your agency should be able to reproduce similar results with said products
- Friction Management integrated as a standard part of railway operations
- Benefits achieved and maintained long-term





Products and Application Methods

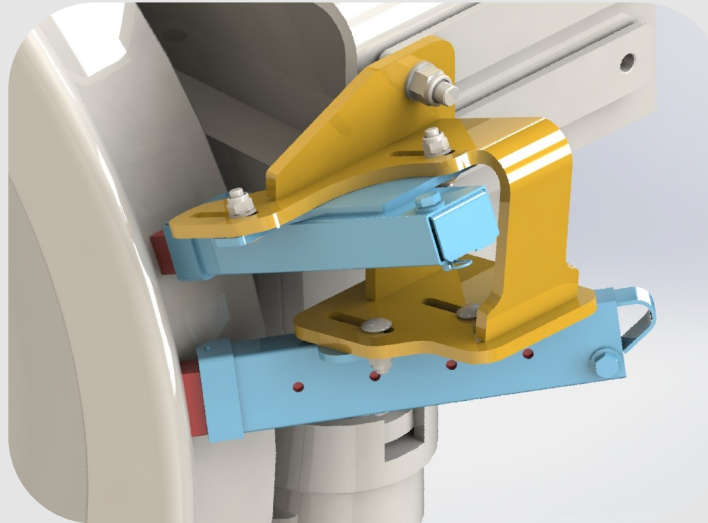
Trackside vs. On-board Application
Top of Rail and Gauge Face Consumables

Application Method Selection

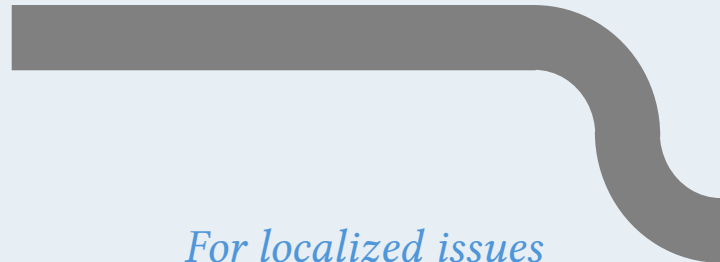
Tracksid



On-Board Solid Sticks



On-Board
Circuit Breaker



For localized issues



For system-wide issues

Consumable Selection

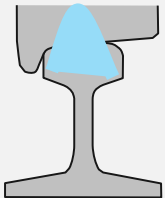
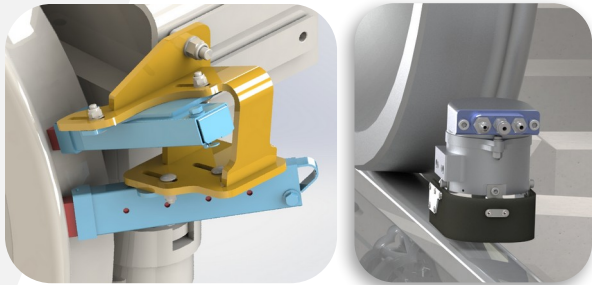
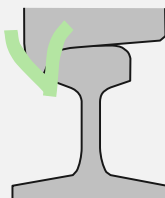
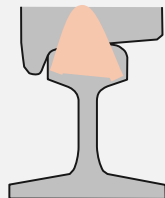


What are the pertinent issues?

Issue	Friction Modifier	Lubricant	Traction Gel
Noise – Squeal (Top of Rail)	✓✓	-	-
Noise – Flanging (Gauge Face / Restraining)	✓✓	✓✓	-
Corrugation	✓✓	-	-
Lateral Forces – Rolling Contact Fatigue	✓✓	-	-
Lateral Forces – Track Structure Degradation	✓✓	-	-
Rail Wear – Gauge Face / Corner	✓	✓✓	-
Rail Wear – Top of Rail	✓✓	-	-
Wheel Wear – Flange (Front or Back)	✓	✓✓	-
Wheel Wear – Tread	✓✓	-	-
Derailment Potential	✓✓	✓✓	-
Energy & Fuel Efficiency	✓	✓	-
Ride Quality & Hunting	✓	-	-
Leaf Fall / Steep Grades	-	-	✓✓

FM Application Matrix



			APPLICATION	
CONSUMABLE		Top-of-Rail / Wheel Tread Friction Modifier	Trackside	On-Board
				
			- Water-based Friction Modifier - Top-of-Rail (TOR) Oil - Oil-based Hybrid	- Solid Stick Friction Modifier - Friction Modifier Spray - TOR Oil-Based Spray
		Gauge Face / RR / Wheel Flange Lubrication	<i>Gauge Face (GF) or Restraining Rail (RR)</i> - Rail Curve Grease	<i>Front or Back of Flange</i> - Solid Stick Lubricant - Oil Spray
		Traction Enhancers	- Traction Gel	- Sand - Traction Gel - Innovative Cleaning Tech

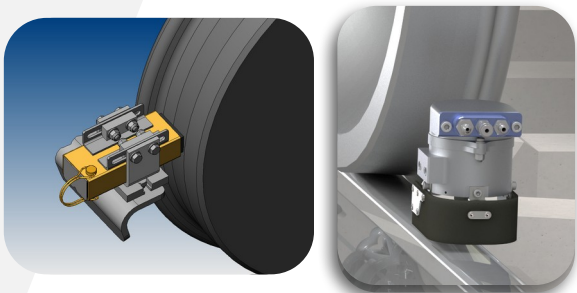
FM Application Matrix

APPLICATION

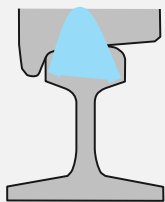
Trackside



On-Board



CONSUMABLE



**Top-of-Rail /
Wheel Tread
Friction Modifier**

- Water-based Friction Modifier
- Top-of-Rail (TOR) Oil
- Oil-based Hybrid

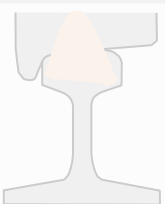
- Solid Stick Friction Modifier
- Friction Modifier Spray
- TOR Oil-Based Spray



**Gauge Face / RR /
Wheel Flange
Lubrication**

- Rail Curve Grease
- Gauge Face (GF) or Restraining Rail (RR)*

- Solid Stick Lubricant
 - Oil Spray
- Front or Back of Flange*



Traction Enhancers

- Traction Gel

- Sand
- Traction Gel
- Innovative Cleaning Tech



How do TOR / Wheel Tread Products Work?

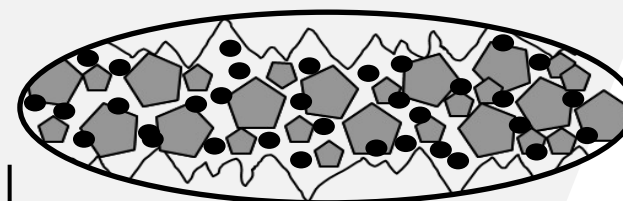


Dry Contact

High
 μ

Friction Modifier Treated Contact

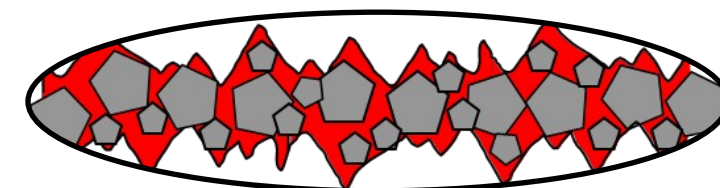
Pliable “soft” particles into wheel-rail interface to counteract brittle “hard” iron oxide particles



TOR Oil Treated Contact

Lubricates surfaces of wheel and rail and requires precise application amounts

Excess application will affect train acceleration and deceleration



Intermediate
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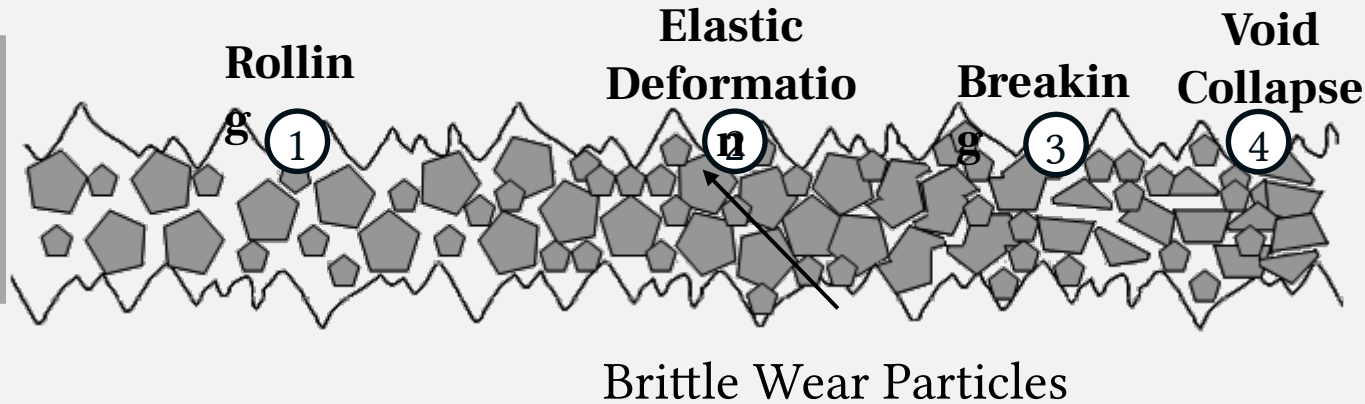


Friction Modifier Treated WRI

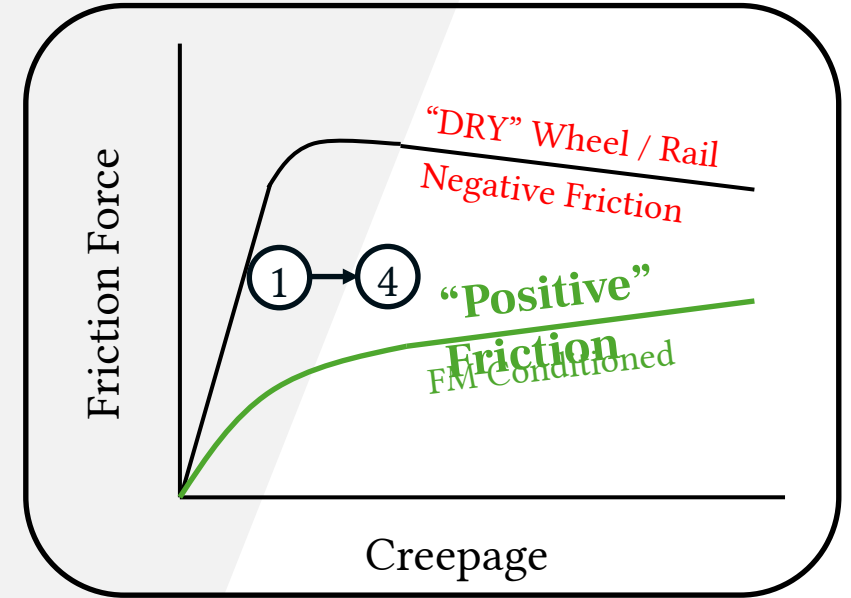
Relative Motion (Accumulating Creepage)



Dry WRI

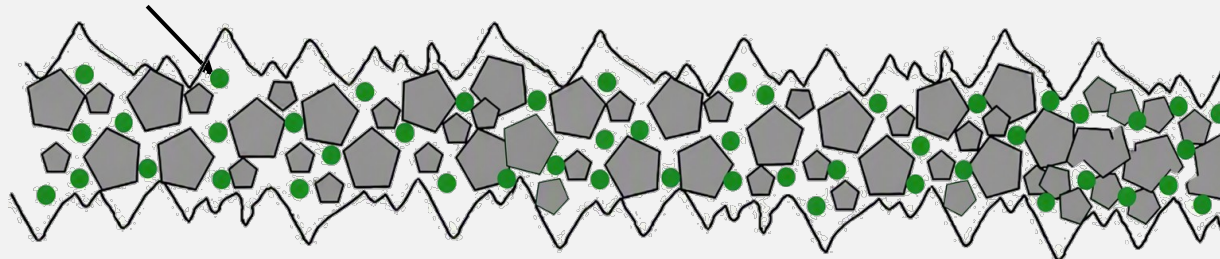


Void collapse directly related to stick-slip oscillations that cause noise, corrugations, etc.



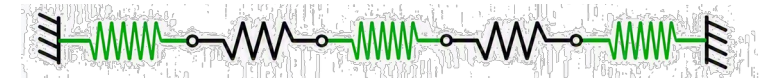
Pliable “Soft” FM Particles

FM Treated WRI



(not to scale)

Reduced wear particle generation



Pliable FM particles mitigates brittle particle breaking and void collapse



TOR Consumable - Key Specifications

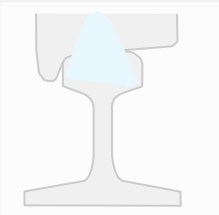
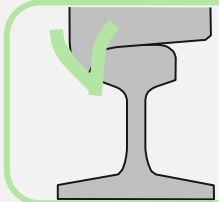
ALL	• Intermediate friction • Positive friction • Good film durability & carry down • Low environmental & health impact • Safe operation
LIQUID	• Effective pumpability & pick-up at operating temperatures • Water wash-off resistance • Product stability
SOLID	• Good mechanical strength & thermal stability • Optimized consumption rate • Interlocking (continuous) design





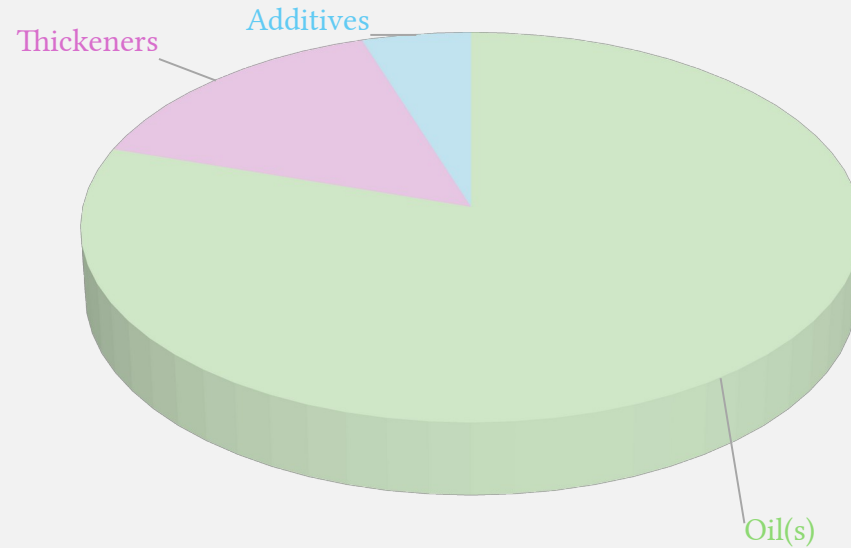
FM Application Matrix



		APPLICATION	
		Trackside	On-Board
CONSUMABLE	 Top-of-Rail / Wheel Tread Friction Modifier	- Water-based Friction Modifier - Top-of-Rail (TOR) Oil - Oil-based Hybrid	- Solid Stick Friction Modifier - Friction Modifier Spray - TOR Oil-Based Spray
	 Gauge Face / RR / Wheel Flange Lubrication	- Rail Curve Grease <i>Gauge Face (GF) or Restraining Rail (RR)</i>	- Solid Stick Lubricant - Oil Spray <i>Front or Back of Flange</i>
	 Traction Enhancers	- Traction Gel	- Sand - Traction Gel - Innovative Cleaning Tech

Basics of Greases

Roughly speaking, a grease is comprised of the following:



There are many different oils, thickeners, and additives!

There are greases designed specifically for railroad operations

Choosing a high-quality grease designed specifically for railroad operations is optimal



Lubricant Consumable - Key Specifications

ALL	• Low coefficient of friction • Good film durability & wear protection • Effective product carry down and transfer • Low environmental & health impact
LIQUID	• Low bar clogging propensity • Good column strength • Optimized pumpability at operating temps • Optimized oil separation • Good adhesion to rail & water wash-off resistance
SOLID	• Good mechanical strength & thermal stability • Optimized consumption rate • Interlocking (continuous) design



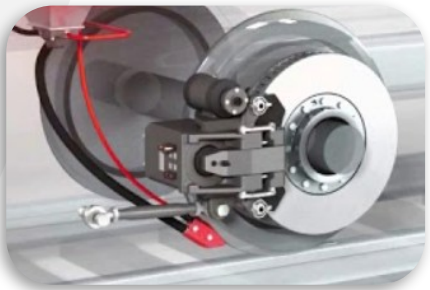
FM Application Matrix

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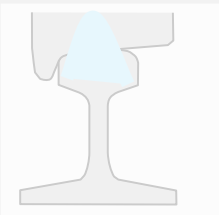
Trackside



On-Board



CONSUMABLE



**Top-of-Rail /
Wheel Tread
Friction Modifier**

- Water-based Friction Modifier
- Top-of-Rail (TOR) Oil
- Oil-based Hybrid

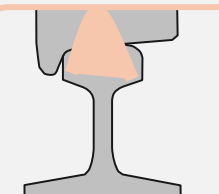
- Solid Stick Friction Modifier
- Friction Modifier Spray
- TOR Oil-Based Spray



**Gauge Face / RR /
Wheel Flange
Lubrication**

- Rail Curve Grease
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- Solid Stick Lubricant
 - Oil Spray
- Front or Back of Flange*



Traction Enhancers

- Traction Gel

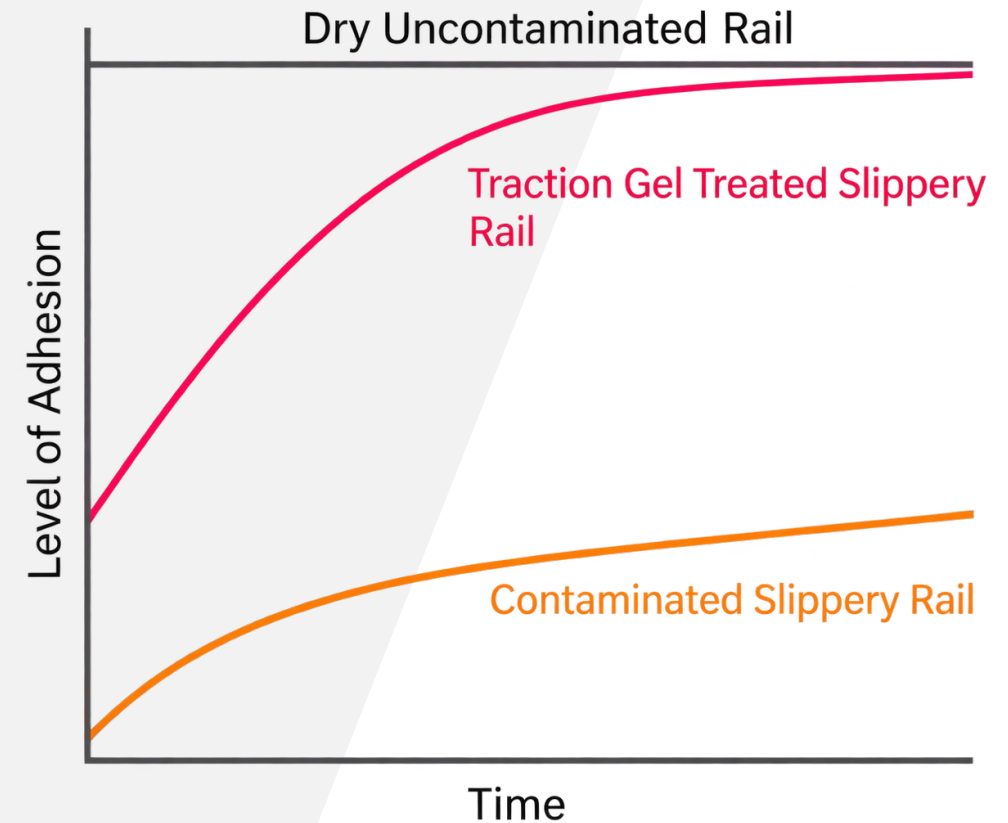
- Sand
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Trackside Traction Enhancer Gel

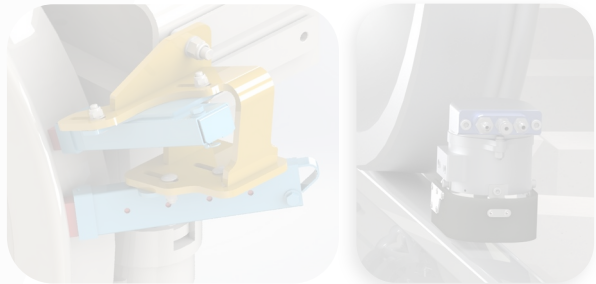
Key Specifications

- Effective traction enhancement
- Effectively breaks down leaf film
- Good film durability & carry down
- Effective pumpability & pick-up at operating temperatures
- Product stability
- Low environmental & health impact



FM Application Matrix

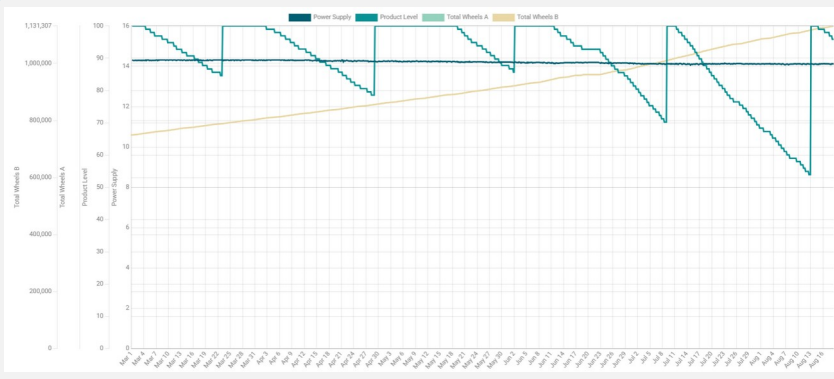
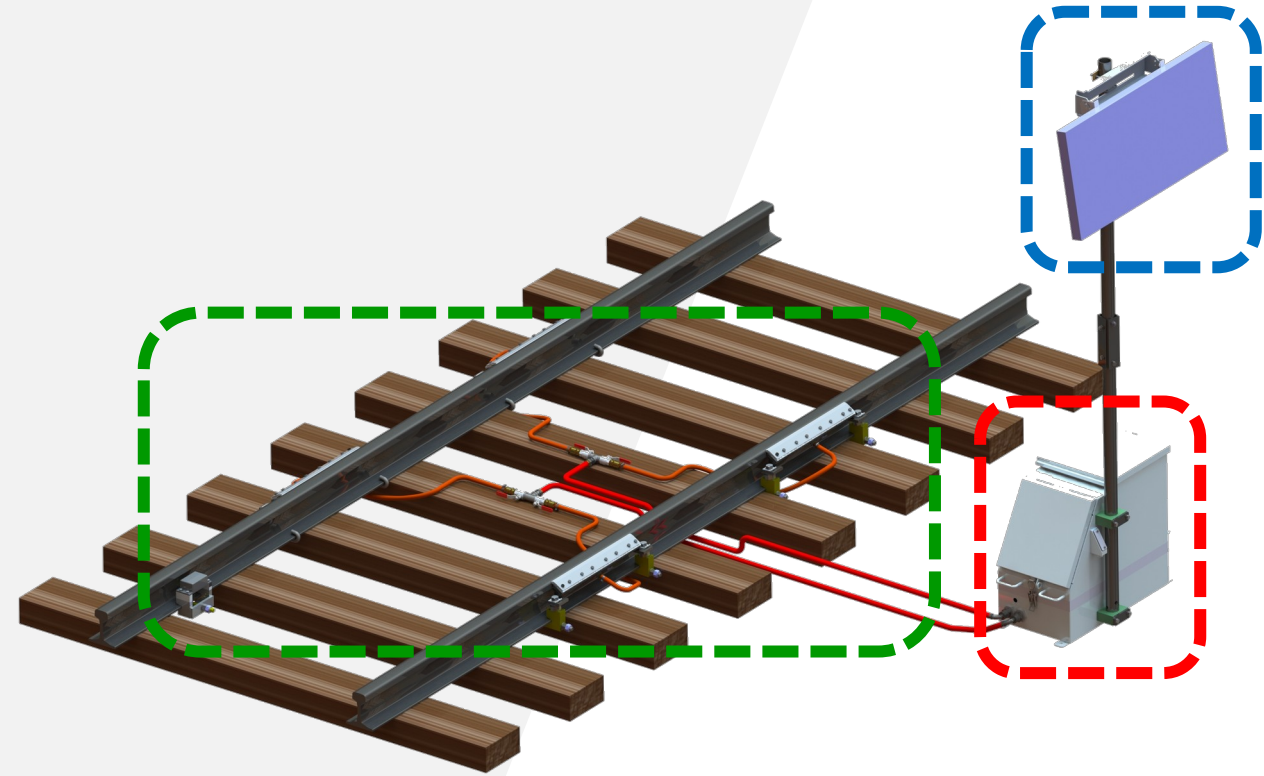


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CONSUMABLE			Trackside	On-Board
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		Traction Enhancers	- Rail Curve Grease <i>Gauge Face (GF) or Restraining Rail (RR)</i>	- Sand - Traction Gel - Innovative Cleaning Tech
			- Rail Curve Grease <i>Gauge Face (GF) or Restraining Rail (RR)</i>	

Trackside Equipment

Key Specifications

- Tank size, shape, and designs
 - Optimize refilling efficiency
- Single or dual track application
- Type and number of applicator bars
- AC or DC Solar power
- Directional application control
- Exposed or embedded rail
- Remote monitoring technologies
 - Available for most trackside equipment
 - Pinpoint and notify on issues, refilling requirements, etc.



Top-of-Rail



Gauge Face



Restraining



Traction Gel

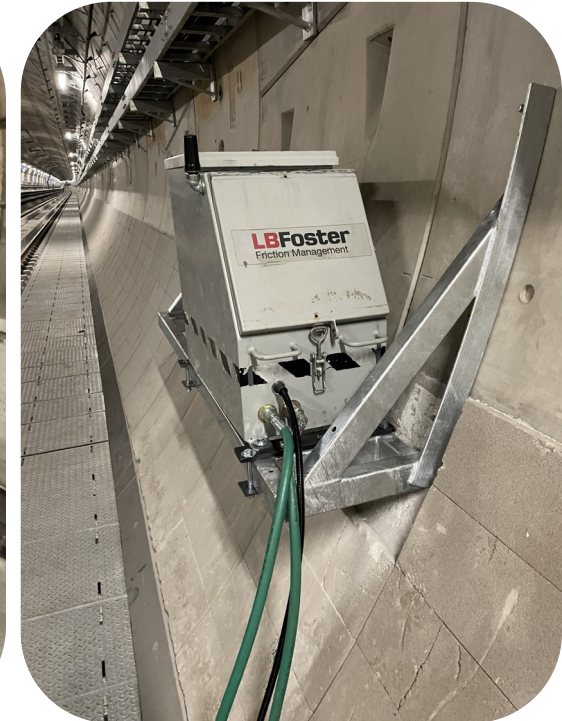
Trackside Placement Considerations

Lubricators are typically placed in tangent track preceding the targeted curve(s) to ensure product is picked up and transferred effectively

- Clearance constraints, especially in subway tunnels and elevated track
- Proximity to power sources
- Ease of access to the unit

For transit lines in the development phase, it is best to create dedicated spaces for lubricators.

Expect noise, wear, and corrugation to occur for any curves less than 350 m / 1050 ft radius!



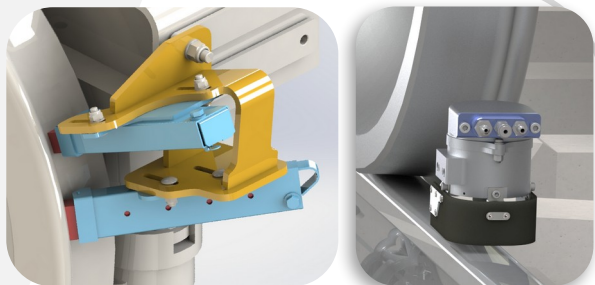
FM Application Matrix

APPLICATION

Trackside



On-Board



CONSUMABLE



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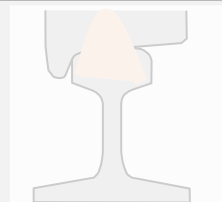
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**Gauge Face / RR /
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Traction Enhancers

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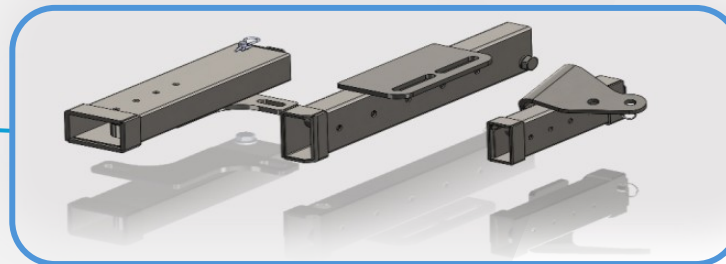
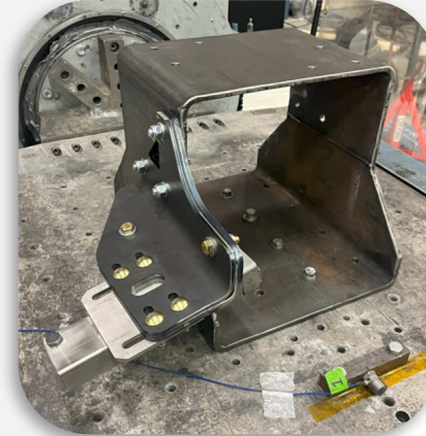
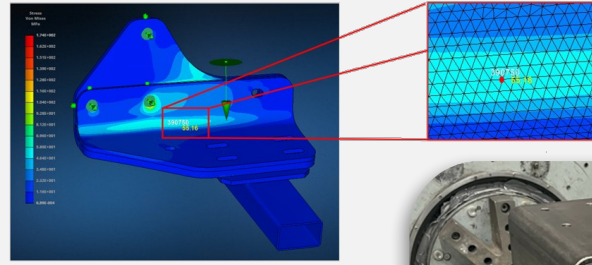
- Sand
- Traction Gel
- Innovative Cleaning Tech



On-Board Equipment

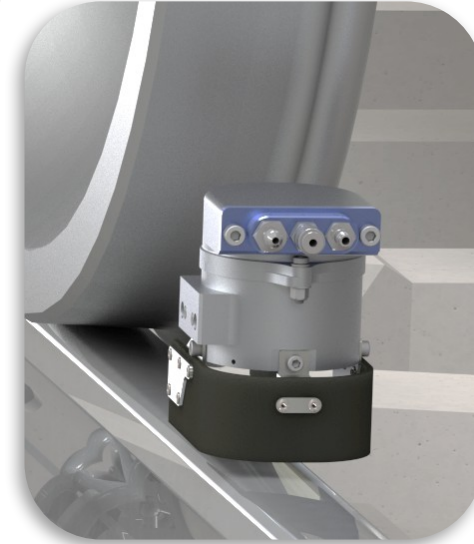
Key Specifications

- Stress testing
 - Finite element analysis
 - Shock & vibration testing
- Integration with specific vehicles
- Adjustability within vehicles
- Compatibility with consumable(s)



Solid Stick

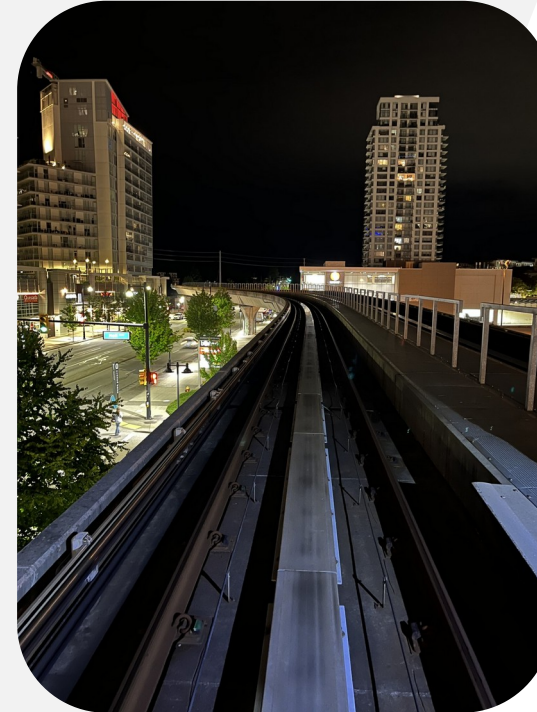
B O S T O



Spray

On-Board Considerations

- On-board equipment required if the following concerns cannot be addressed by trackside equipment:
 - Number of trackside units too high to realistically maintain
 - Insufficient clearance or space restrictions for trackside units, e.g., abundance of elevated track
- Requires a sufficient percentage of fleet to be outfitted with on-board equipment; applicable to both solid sticks and spray systems



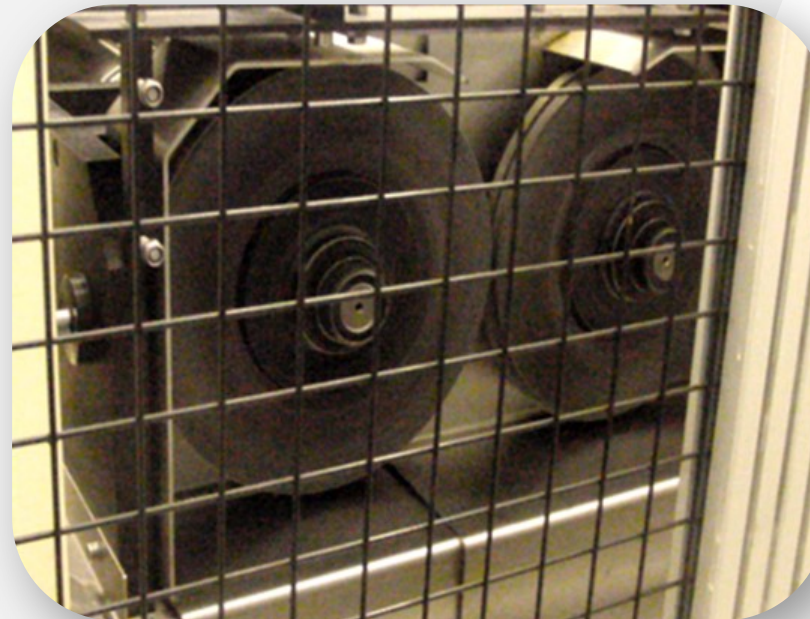
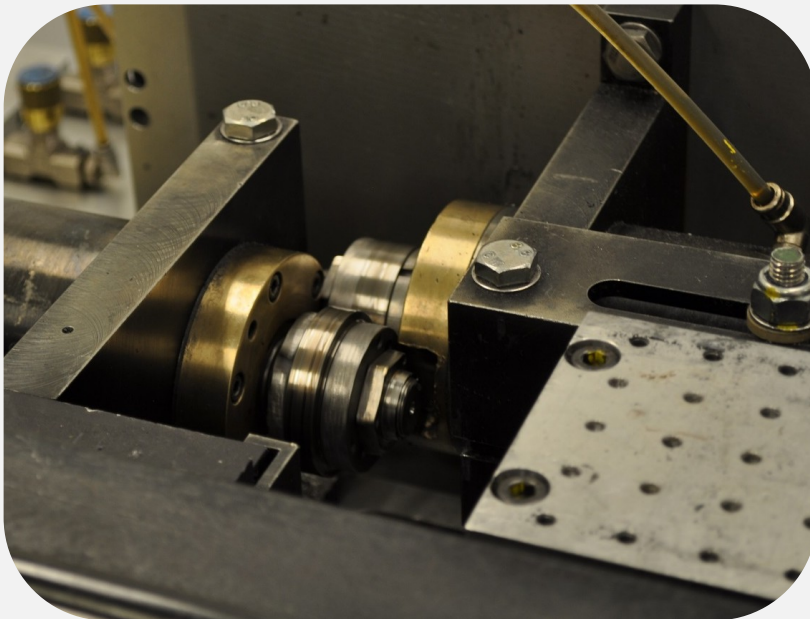


Validation

Field and Laboratory Measurements

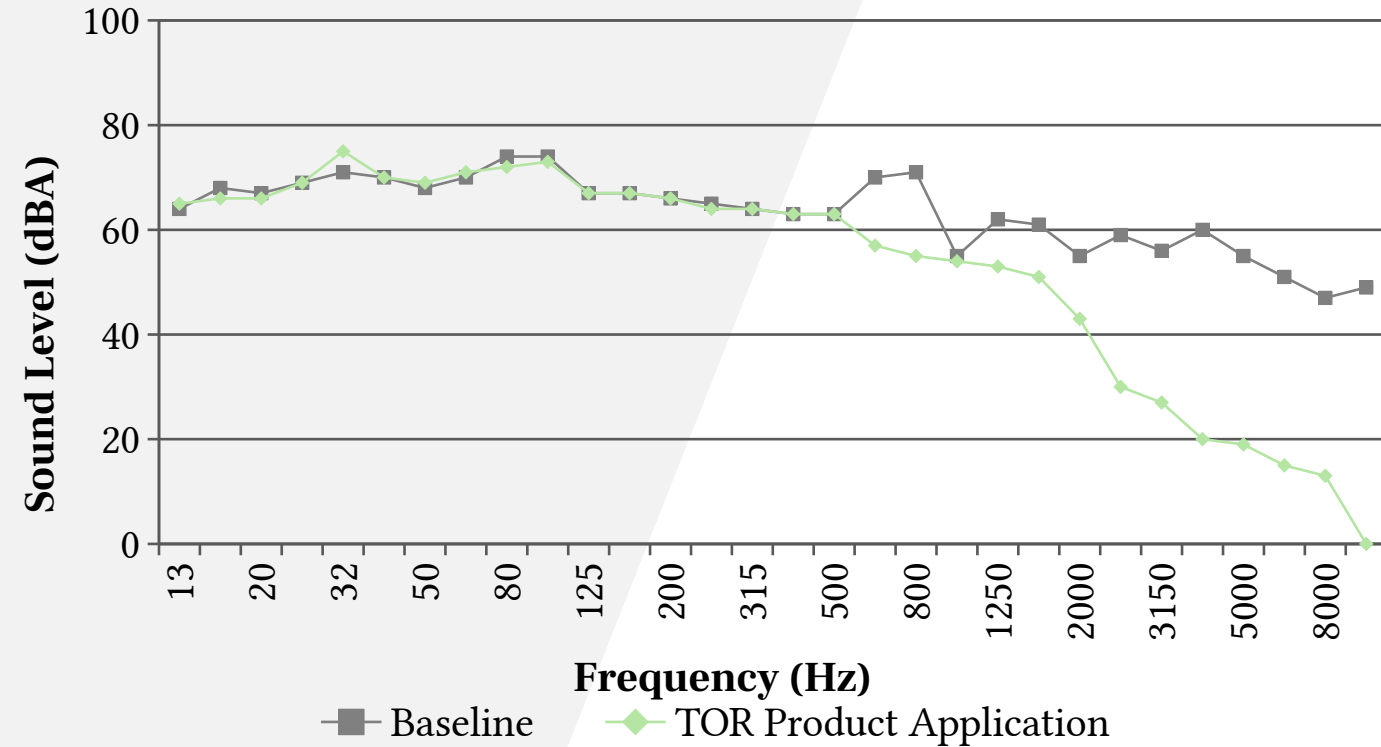
Laboratory Validation

- Prototyping
 - Using laboratory tests directly relatable to wheel/rail interface
 - Tests correlate well to field and are getting refined and implemented into EN standards
- Field Sample Validations



Noise Measurements

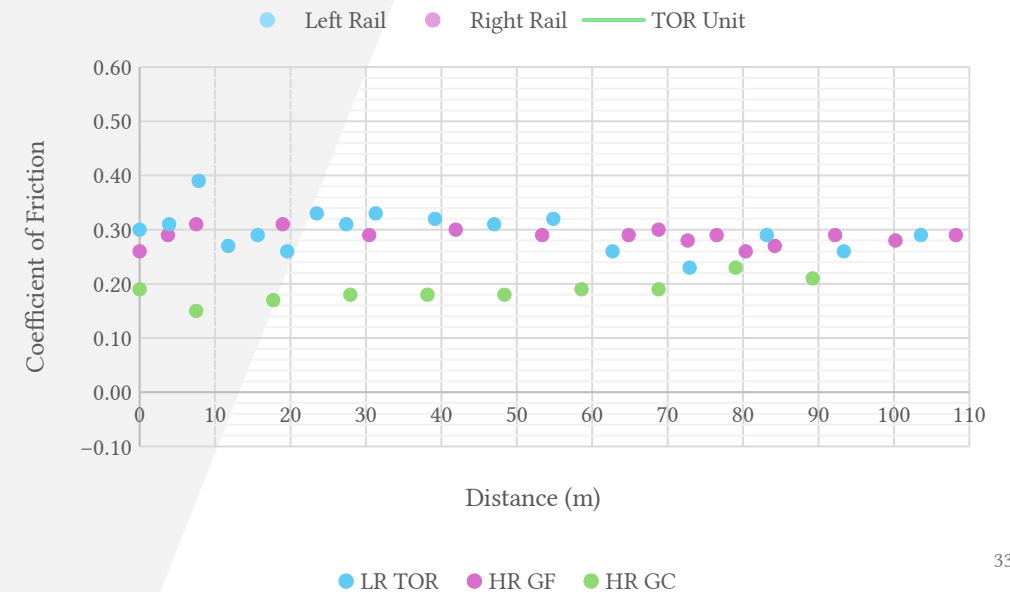
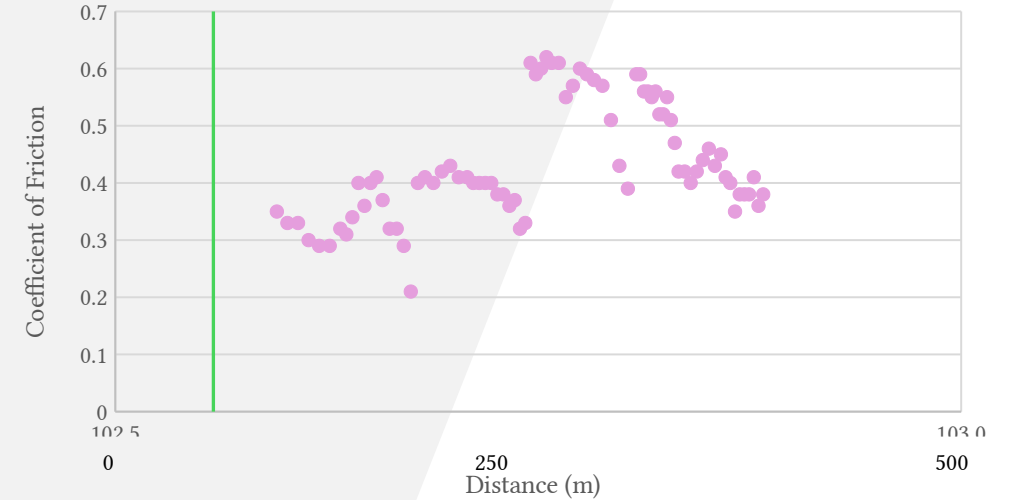
Noise Type	Frequency range (Hz)
Rolling	30 – 2500
Rumble (including corrugations)	200 – 1000
Flat spots	50 – 250 (speed dependant)
Ground borne vibrations	30 – 200
Top-of-rail squeal	1000 – 5000
Flanging noise	5000 – 10 000



Carry Distance Measurements

Tribometer Test

- Measure friction levels near lubrication sites
- Can be used to measure carry characteristics of both TOR and GF products



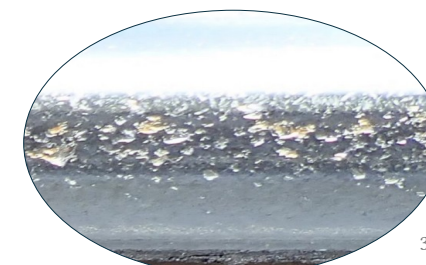
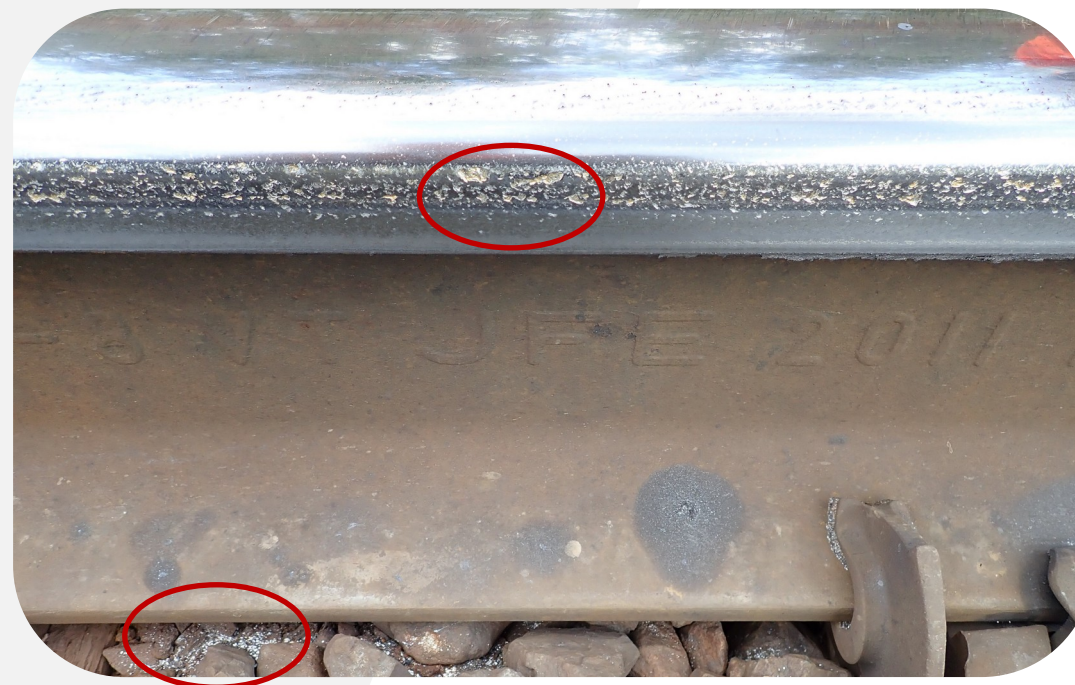
Carry Distance Measurements

- Visual inspections for qualitative assessment of grease
- Not applicable for TOR products!

Good Grease Coverage

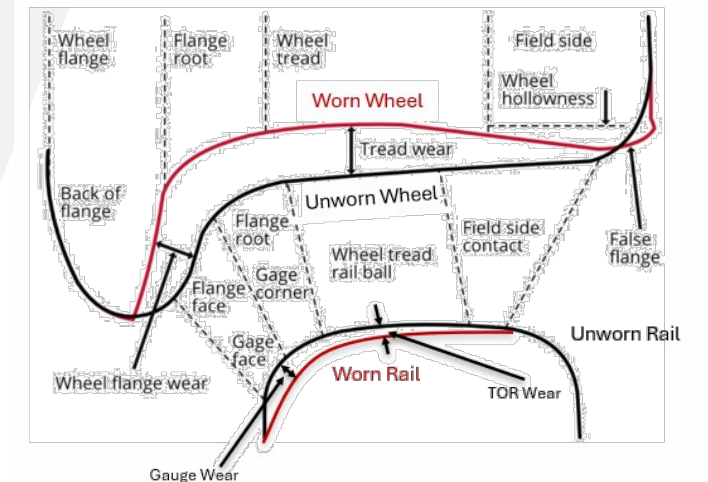


Poor Grease Coverage

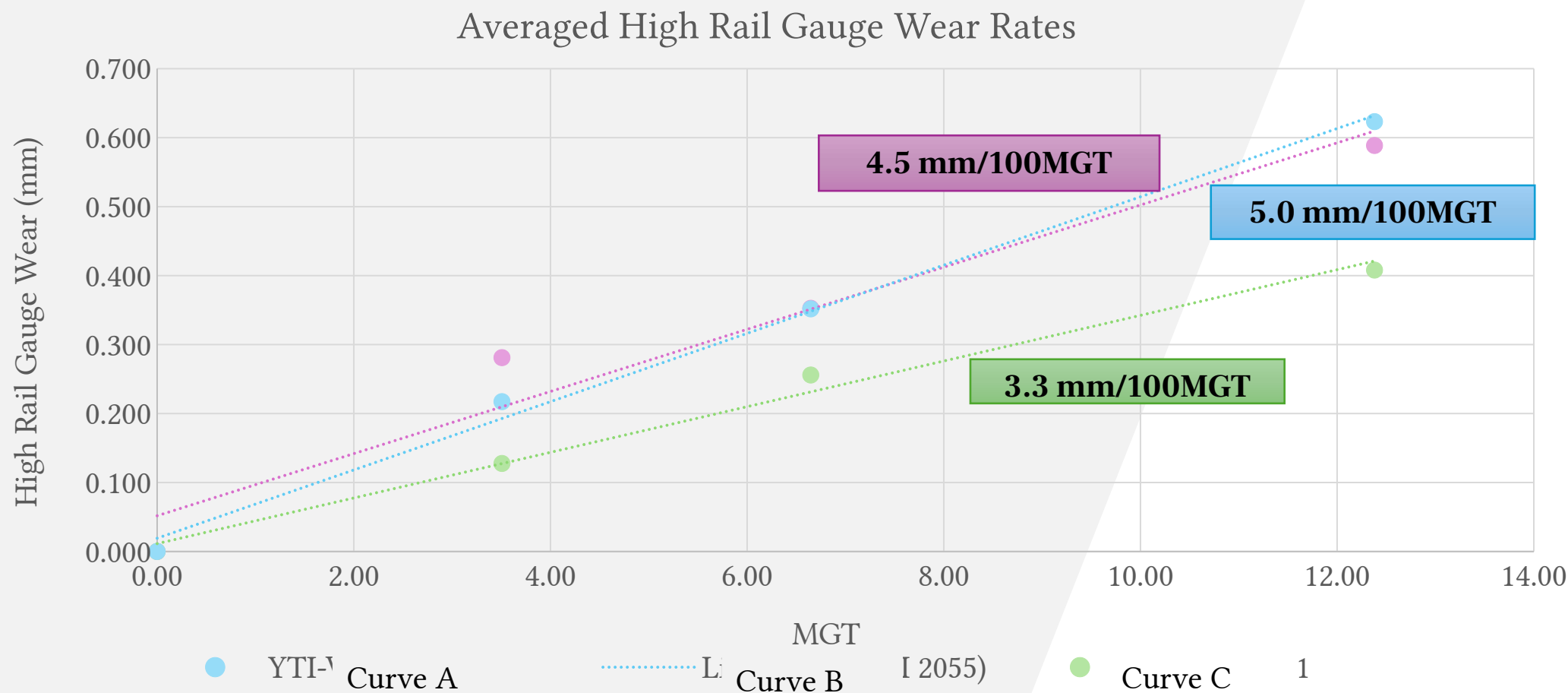


Rail & Wheel Wear Measurements

- MiniProf gauges
- Onboard track measuring vehicles



Rail Wear Example



Corrugation Measurements

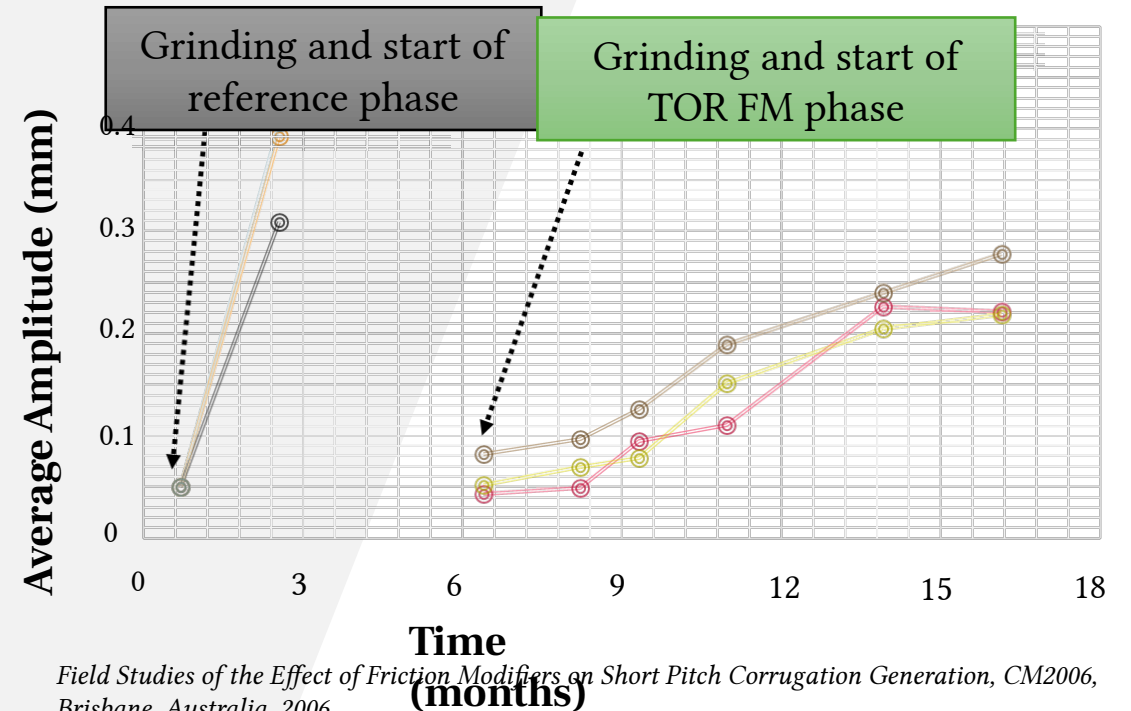
- Measure corrugations/roughness with **C**orrugation **A**nalysis **T**rolley
- Must grind or mill rails prior to running a corrugation trial or measure corrugation abatement with TOR FM



Reference Phase

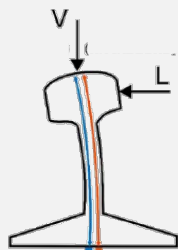


TOR Phase

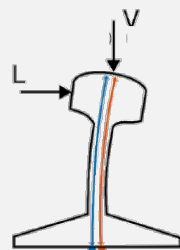


Lateral Force Measurements

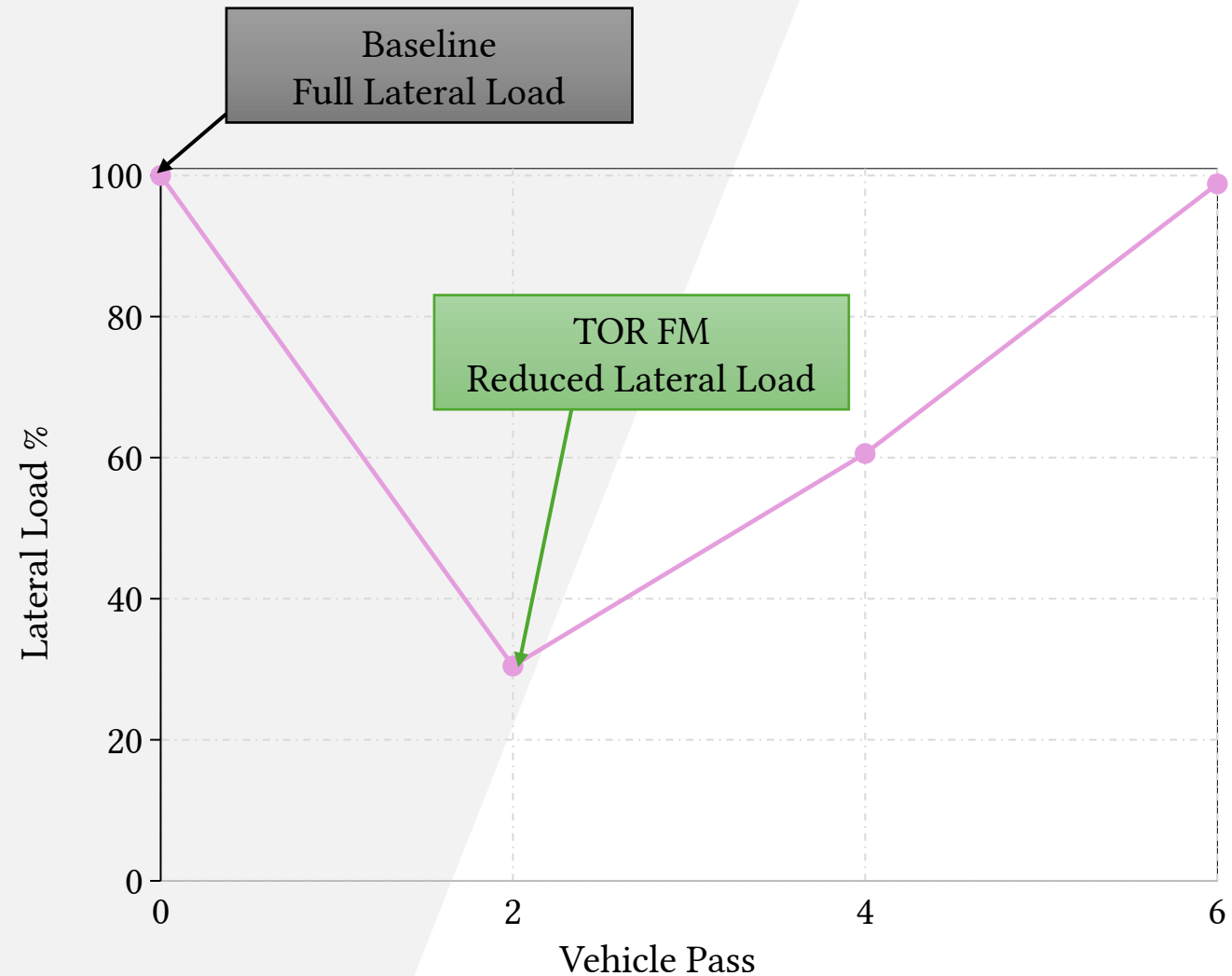
- Measure reduction in lateral loading with ultrasonic equipment
- Example of single spray from onboard TOR FM applied after baseline measurement
- Measured how many train passes required to restore lateral load to baseline conditions
- Transit train run back and forth at test curve and lateral loading measured (inbound direction only)



High Rail

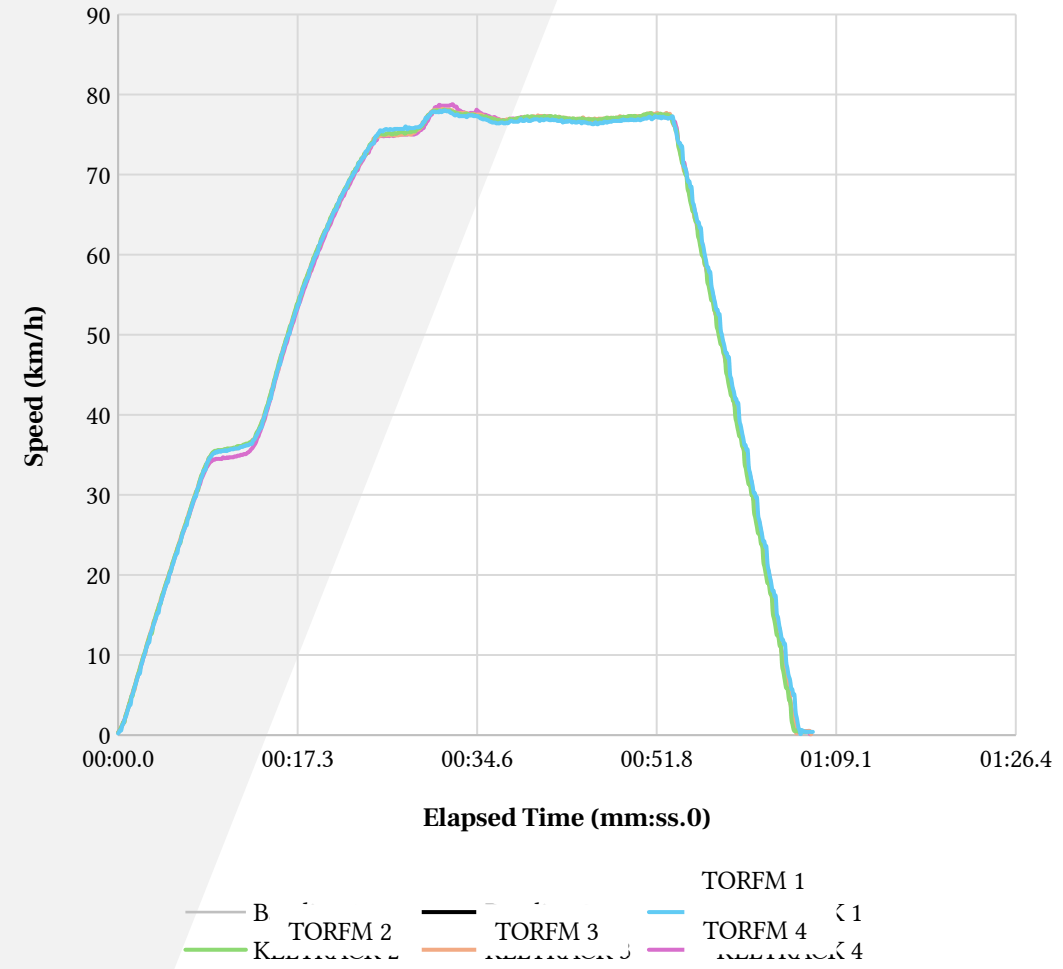
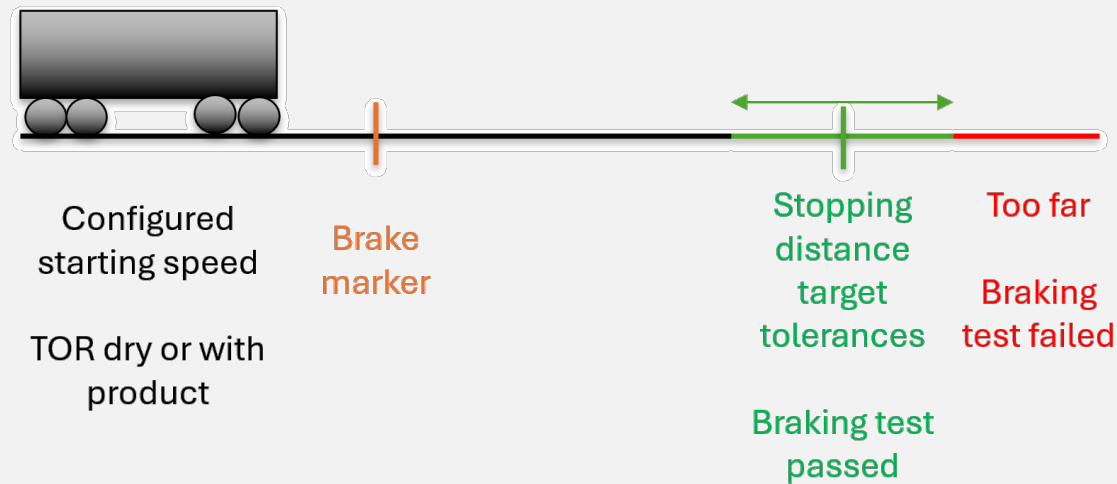


Low Rail



Safety - Braking and Traction Tests

- Common agency approval requirement
- Braking must pass agency's safe criteria
 - Configured starting speeds
 - Stopping distances and deceleration rates measured
 - Ensure braking does not cause wheel slide issues



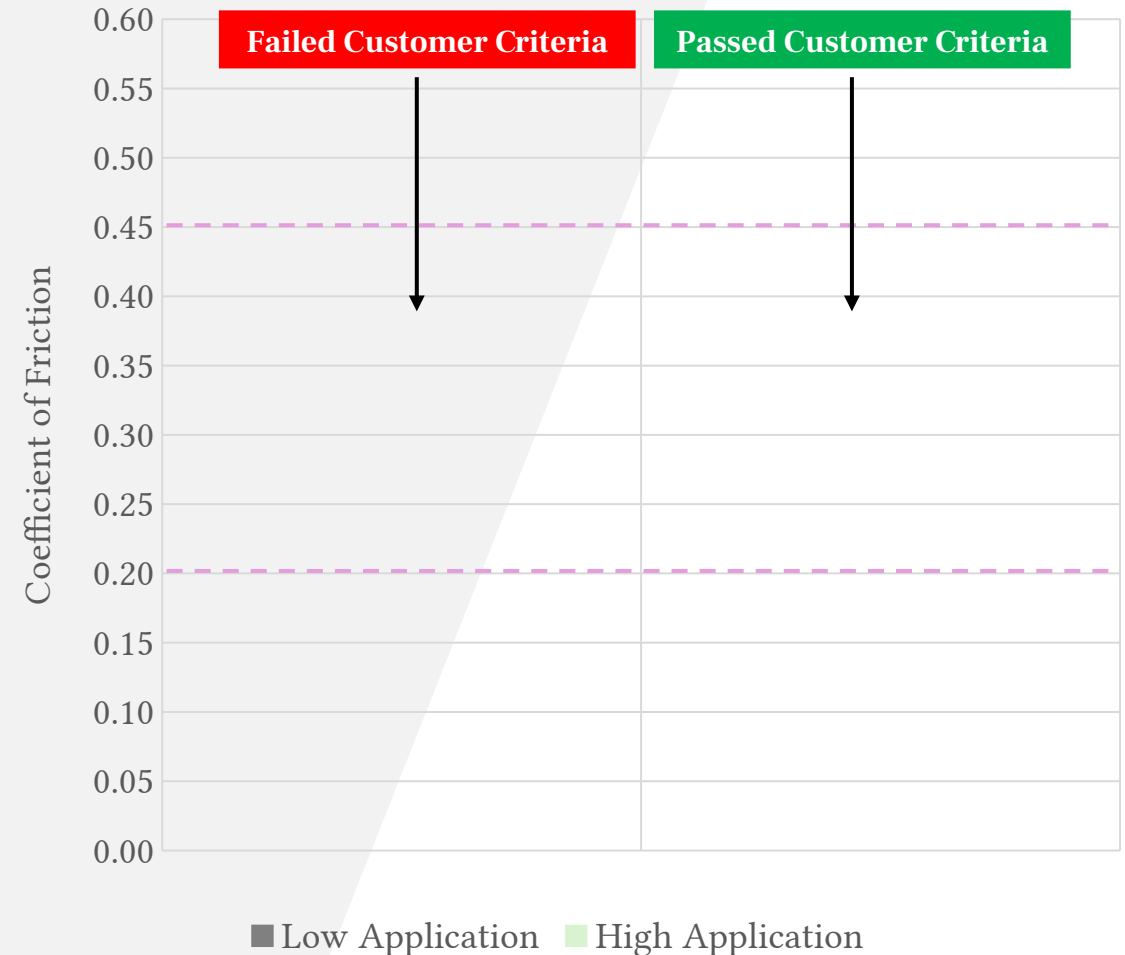
Application Rates and Friction Levels

Correct application rates are critical for FM to function properly

- Underapplication of product provides reduced or potentially negligible FM benefits
- Validate output rates with volumetric outputs
- Equipment in industry can simulate train movement

Example of optimizing application rates

- Lower application rates failed customer criteria
- Higher application rates passed customer criteria





Maintenance

Sustaining a Friction Management Program

Maintenance & Upkeep

Selecting the right equipment and consumable solution is the first step, but need to sustain the FM program

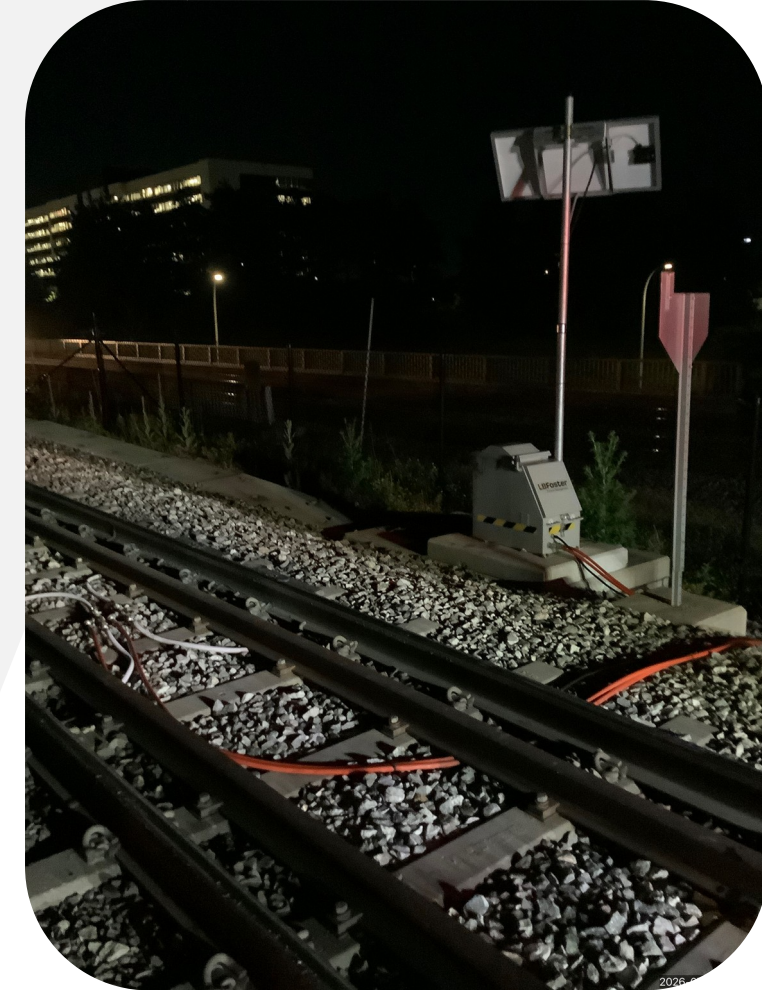
Those who do Friction Management well:

- Know why they are doing it
- Have a leader who is directly responsible
- Know what equipment they have and where it is
- Dedicate trained individuals specifically to this task
- Incorporate maintenance into their processes
- Manage an inventory of spare parts & consumables
- Ensure Friction Management is specified in capital & operating programs and budgets

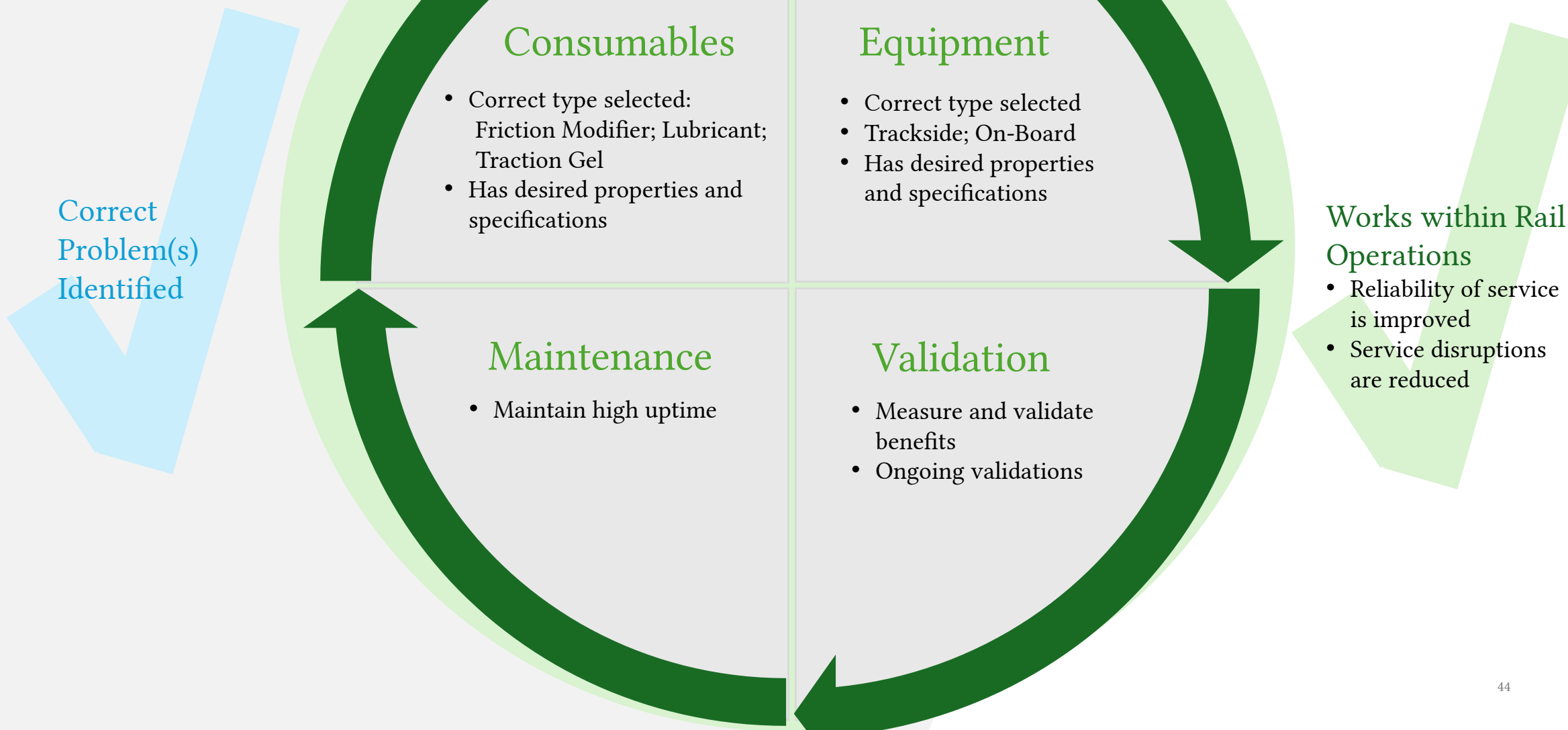


Maintenance Challenges

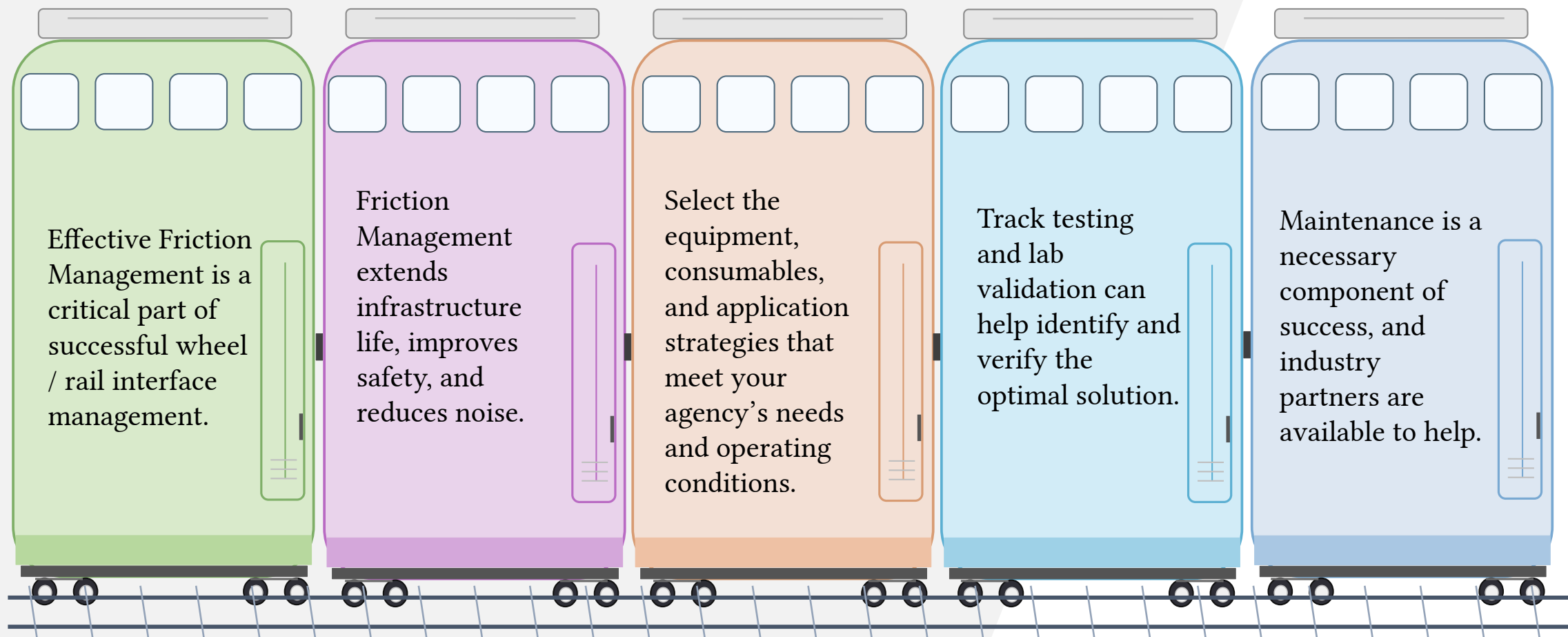
- 1) Very limited maintenance windows: strategic planning is key!
 - Plan out the work ahead of track time
 - Efficient transport: use hi-rail trucks, work trains, speeders, etc. to transport and service equipment and consumables
 - Maintenance scheduling: use remote monitoring to prioritize locations
- 2) Lubrication often is one of many responsibilities of engineering and/or maintenance staff
 - Friction Management must be included as part of maintenance program
 - If resource constraints prevent Friction Management from properly being done, outsource work to industry experts



FM Program Checklist



Summary





Thank You!